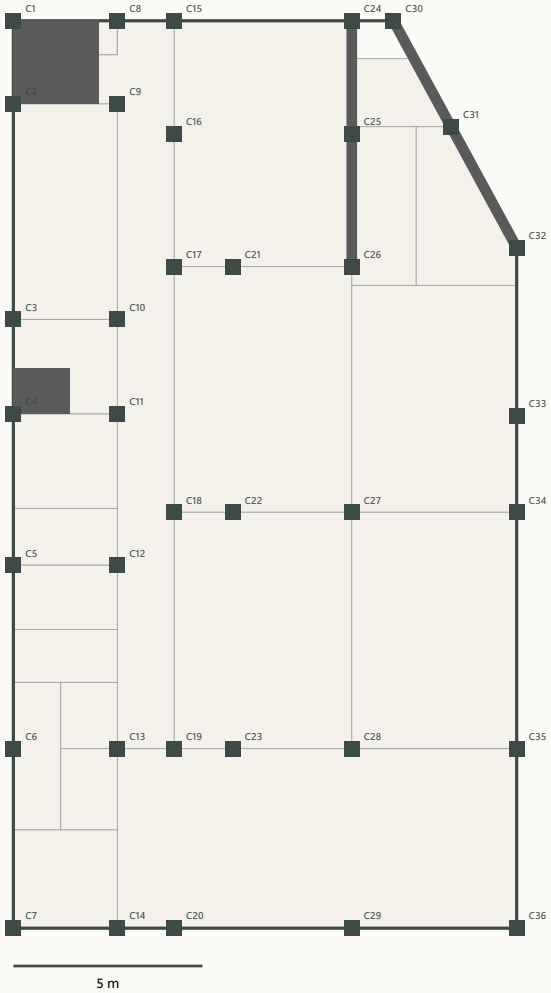
Preliminary structural study · Taif / Palm House V6.3

architectural coordination study · every size ASSUMED, to be calculated by the structural engineer · beam depths are TOTAL depths including the slab · CSV coordinates, north up

ALL CHECKS PASSED means the drawings are consistent; it does not mean strength, deflection or vibration were calculated.

Ground floor · columns (■ in the walls, ● colonnade, all to the roof: X8), RC boxes (dark), tie beams on the column lines



The idea: a simple RC skeleton, walls on walls

- A **reinforced-concrete frame** (columns, beams, slabs) with **block infill**, as Saudi contractors build every villa. Columns stand on the existing wall lines, hidden in the walls; **every column runs from its footing to the roof** (the three colonnade columns too, since X8: they carry the edge beam B3 and the roof edge beam R11). No transfer beam carries a column.
- **Spans 2.2–6.5 m.** The only spans above 6 m are on the courtyard lines (x 4.25, 5.8, 8.95: 6.5 and 6.25 m), because the courtyard and the double-height living must stay free of columns; they get drop beams ≈ 0.50–0.55 total depth (0.30–0.35 below the slab).
- **Slabs:** solid 0.20 m where spans are small (west band, riwaq, projection, roofs of 4.7 m), hollow-block **ribbed 0.30 m** (25+5) for the 4.36–4.75 m bays of the east band and the men's majlis roof. The double-height living's roof is a **flat solid slab 0.22 m spanning 4.7 m east–west** between the two long walls: **no beam crosses the void** (§ options; a common span for a 0.22 solid slab, not calculated).
- **Foundations (assumed):** isolated pad footings ≈ 1.5 × 1.5 m under the columns, strip footings under the RC boxes, RC tie beams on the column lines at ground level; the lift pit 1.2 m deep; the air duct (studies/air_system.typ v2, route A): a small dry sealed box 0.9 × 0.9 (0.6 × 0.6 clear, assumed) leaving the tower base pit westward and running under the women's majlis to a chamber under the courtyard floor: top −1.00, just under the tie beams, clear of the pads except at its two ends. **No soil report yet** (X9).

Balanced stiffness · a proposal, NOT calculated

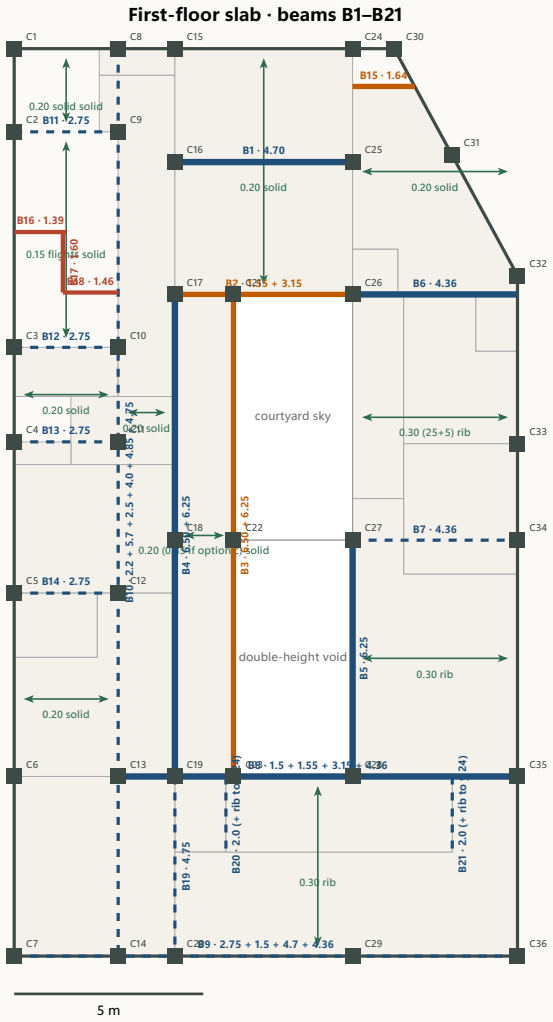
- **West:** the lift shaft (W1, RC walls on 3 sides), the west shaft (W2) and the stair walls make a stiff core at x 0–2.75 (its stiffness is not calculated here).
- **East, to match it:** the solid laundry / maid wall on x 8.95 (W3, no openings, 6.5 m) and the cut wall (W4) are proposed as RC walls 0.20 m (W5, the kitchen's north wall, is dropped in V6.4: that wall is now a partition at y 7.0 and the y 6.5 line is the beam B6), intended as an east box of a similar order of stiffness at a similar distance from the centre, so that the centre of stiffness stays near the centre of mass. **Not calculated:** the engineer models the lateral system (frame + walls) and confirms it.
- **Why:** the house is a long pentagon (13.3 × 24 m); with stiff elements only on the west, wind and earthquake would tend to twist it about the west core. The seismic design category for Taif and the wind case are to be taken from SBC 301 by the engineer; not assessed here.

Column lines and spans

Line	Columns at y (or x)	Spans (m)
x 0 and x 2.75	y 0 · 2.2 · 7.9 · 10.4 · 14.4 · 19.25 · 24	2.2 · 5.7 · 2.5 · 4.0 · 4.85 · 4.75
x 4.25 and x 8.95	y 0 · 3.0 · 6.5 · 13.0 · 19.25 · 24	3.0 · 3.5 · 6.5 · 6.25 · 4.75
x 5.8 (colonnade, to the FF)	y 6.5 · 13.0 · 19.25	6.5 · 6.25
cut wall	(10.051, 0) · (11.572, 2.8) · (13.311, 6.0)	3.2 · 3.6
x 13.311	y 6.0 · 10.45 · 13.0 · 19.25 · 24	4.45 · 2.55 · 6.25 · 4.75
y 0 / y 24 (cross)	x 0 · 2.75 · 4.25 · 8.95 · 13.311 (cut corner at 10.051)	2.75 · 1.5 · 4.7 · 4.36 (1.1)
y 6.5 · 13.0 · 19.25 (cross)	x 4.25 · 5.8 · 8.95 · 13.311 (+ 2.75 at y 19.25)	1.55 · 3.15 · 4.36

First-floor slab (+3.40 … +3.60) · beams and slab directions

blue = drop beams · dashed = hidden beams (slab depth) · orange = edge beams of the openings · red = stair trimmers · green arrows = slab span · sizes width × depth, assumed



Beam	Span (m)	Size, assumed	Carries / note
B1	4.70	0.25×0.40 total = 0.20 below the slab, soffit +3.20; hidden by the women's majlis gypsum ceiling (≈ +3.05, owner decision)	FF kids' room / family living wall (y 3.0, nothing below) + the women's majlis roof
B2	1.55 + 3.15	0.25×0.40 hidden / drop	FF slab edge on the courtyard (the family living's steel window W2 stands on it, V6.4); over the women's majlis south glazing
B3	6.50 + 6.25	0.25×0.50 total · UPTURNED (owner's preference: 0.30 above the slab = the sill of the gallery window W1, V6.4) or drop (0.30 below; the engineer decides) · § options	the 1.35 m projection and the gallery, on C21–C23
B4	6.50 + 6.25	0.25×0.50 total = 0.30 below the slab, soffit +3.10	FF west rooms' east wall (over the riwaq glazing and the living's entrance arch; on the solid TV wall from y 14.5, V6.3) + the projection back span
B5	6.25	0.25×0.55 total = 0.35 below the slab, soffit +3.05 (or 0.60×0.40 wide-shallow, soffit +3.20)	FF suite bedroom west wall + half the dining roof; over the open ceiling-height change
B6	4.36	0.25×0.45 total = 0.25 below the slab, soffit +3.15 (V6.4; was hidden 0.30 in the wall)	V6.4: no wall below any more (the kitchen's north wall moved to y 7.0 as a non-load-bearing partition): a drop beam from C26 to the east wall inside the laundry / maid room ceiling, hidden by the gypsum ceiling (≈ +3.0); carries the FF bedroom 1 / suite wall above
B7	4.36	hidden 0.30	on the kitchen / dining wall; the suite dressing floor spans over it
B8	1.5 + 1.55 + 3.15 + 4.36	0.25×0.45 total, 0.25 above the wall	on the men's majlis north wall; carries the FF bedroom 3 / office wall, the suite court north wall, the 7 m living south wall and the treadmill arm
B9	2.75 + 1.5 + 4.7 + 4.36	hidden in the 0.25 wall / 0.25×0.45	south wall; men's majlis roof support
B10	2.2 + 5.7 + 2.5 + 4.0 + 4.85 + 4.75	hidden 0.20–0.30; 0.25×0.40 at the 5.7 m stair bay	riwaq line: west band slabs and the riwaq slab
B11	2.75	hidden 0.20	lift / stair wall
B12	2.75	hidden 0.20	stair / utility wall · FF landing
B13	2.75	hidden 0.20	utility / hospitality kitchen wall
B14	2.75	hidden 0.20	basins / family WC wall · FF bedroom 2 / bath 3 wall
B15	1.64	0.20×0.35 edge	air tower opening (the tower rises through both slabs)
B16	1.39	0.20×0.40 total, soffit +3.20	stair opening edge over flight 1 (headroom 2.22 m at tread 6, to the beam soffit); its east end on B17 (beam on beam: normal at a stair well)
B17	1.60	0.20×0.40 total, soffit +3.20	stair opening, on flight 1's side of the split (x 1.19–1.39), never over flight 2; ends on B16 and B18
B18	1.46	0.20×0.40 total, soffit +3.20	stair opening, the arrival edge (top riser line y 6.35); to the riwaq line beam B10
B19	4.75	hidden rib 0.30 (in the ribbed majlis roof)	FF office / gym route wall, nothing below; a thickened rib spanning y 19.25 → 24
B20	2.0 (+ rib to y 24)	hidden rib 0.30	FF gym route / suite court wall, nothing below
B21	2.0 (+ rib to y 24)	hidden rib 0.30	the treadmill arm's west wall, nothing below; treadmill live + dynamic load

First-floor walls and what carries them

On a wall below (block infill under a hidden beam)

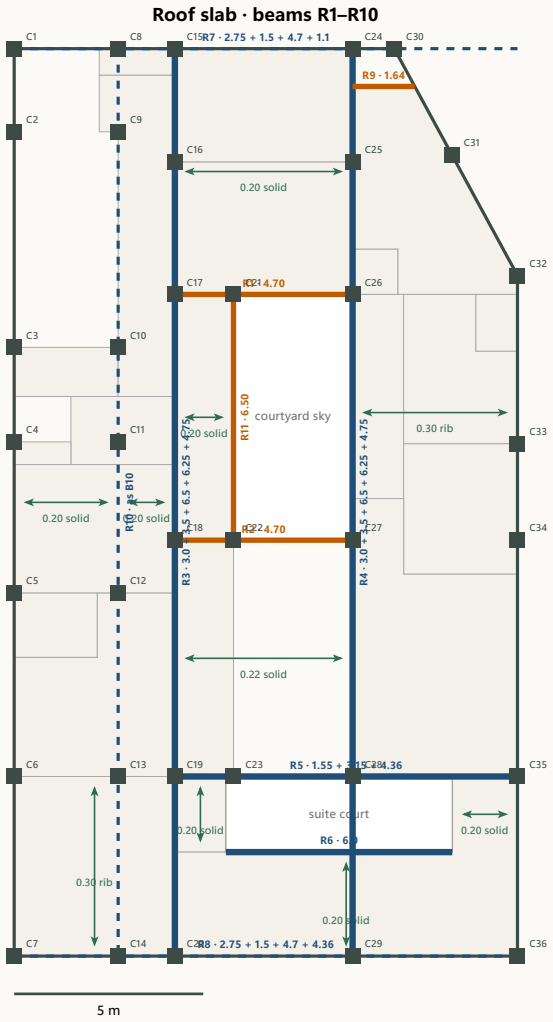
x 0, x 13.311, y 0, y 24, the cut wall (outer walls); x 2.75 at the lift and landing; x 2.25 lift; y 2.2, 7.9, 10.4 (shaft), 14.4, 16.1 (bath 3 = GF family WC), 19.25 (majlis north wall), x 1.5 shaft; x 8.95 y 6.5–13 (courtyard east wall); y 13.0 line inside the suite (kitchen / dining wall)

Nothing below → beam or light partition

y 3.0 kids' / family living → **B1**; **x 4.25 y 13–19.25** bedrooms' east wall over the riwaq glazing and the entrance arch (on the solid TV wall from y 14.5) → **B4**; **x 8.95 y 13–19.25** suite bedroom west wall over the open ceiling change → **B5**; **y 6.5 x 8.95–13.311** bedroom 1 / suite wall over the maid wing (the kitchen's north wall moved to y 7.0, V6.4) → **B6** drop beam; **x 4.25 y 19.25–24** office → **B19** rib; **x 5.6** gym route → **B20**; **x 11.586** treadmill arm → **B21**; **y 21.25** gym north wall (7.7 m, east–west across the ribs) → **light partition on the slab** (X6); suite walls x 10.3, y 10.45, x 12.211, y 8.0, y 11.9, y 13.9 over the kitchen → light partitions (100 mm block or dry wall) on the ribbed slab, or hidden beams under the bath walls; y 11.0 (bath / niche / bedroom 2) and y 9.2 x 1.5–4.25 (family bath north) → light partitions; x 2.2 bath 3 / bedroom 3 entry → light; y 5.3 and x 10.15 (bedroom 1 / family link) → light; linen front y 0.7 → joinery

Roof slab (+7.00 … +7.20) · beams and slab directions

the courtyard sky and the suite court are holes; the double-height living is roofed by a flat slab with no beam in the void; the colonnade C21–C23 continues to the roof (X8), the gallery strip roof on R11



Beam	Span (m)	Size, assumed	Carries / note
R1	4.70	0.25×0.45 drop	courtyard sky hole, north edge (over the FF family living balustrade line)
R2	4.70	0.25×0.45 drop	courtyard sky hole, south edge = the clean double-height roof's north edge
R3	3.0 + 3.5 + 6.5 + 6.25 + 4.75	0.25×0.50 total, 0.30 above the FF wall	on the FF west rooms' east wall / office wall; the gallery strip roof spans 1.35 m from it to R11 on the colonnade (X8)
R11	6.50	0.25×0.45 total, soffit +6.75	roof edge of the courtyard sky over the gallery strip, on C21–C22 (X8, owner: the colonnade continues to the roof, no cantilever)
R4	3.0 + 3.5 + 6.5 + 6.25 + 4.75	0.25×0.50 total, 0.30 above the FF wall	on the FF suite west wall; the double-height roof spans to it
R5	1.55 + 3.15 + 4.36	0.25×0.50 total, soffit +6.70	over the living's south wall (clerestory head +6.00, 0.70 m below the soffit) and the suite court north wall
R6	6.0	0.25×0.50 total, soffit +6.70, DROP (not upturned: winter light, X4)	suite court hole, south edge = gym roof north edge; on the FF walls x 5.6 and x 11.586: proposed load path, NOT verified (X5)
R7	2.75 + 1.5 + 4.7 + 1.1	hidden in the 0.25 wall	north wall
R8	2.75 + 1.5 + 4.7 + 4.36	hidden in the 0.25 wall	south wall
R9	1.64	0.20×0.35 edge	air tower head opening
R10	as B10	hidden 0.20–0.30	riwaq line

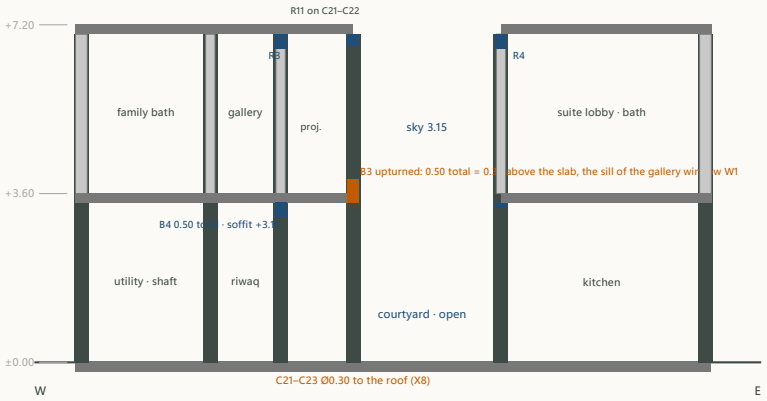
Slab bays (first floor and roof)

Level	Bay (x0 y0 – x1 y1)	Type	Span	Note
FF	0 0 – 2.75 2.2	lift: FF slab opening (the lift is installed at construction, V6.3) 0.20 solid	NS	lift
FF	0 2.2 – 2.75 7.9	stair: L-shaped opening + flight slabs 0.15 flights	NS	opening x 0.125–1.39 y 2.30–4.75 + x 1.39–2.65 y 2.30–6.35 (about 8.2 m2), see the stair page
FF	0 7.9 – 2.75 10.4	solid 0.20	EW	utility; shaft opening 1.5×1.2 framed
FF	0 10.4 – 2.75 24	solid 0.20	EW	west wash block, 2.75 m bays
FF	2.75 0 – 4.25 19.25	solid 0.20	EW	riwaq, 1.5 m
FF	4.25 0 – 8.95 6.5	solid 0.20	NS	women's majlis roof: 3.0 + 3.5 m to beam B1
FF	4.25 6.5 – 5.8 19.25	solid, the projection 0.20 (0.35 if option c)	EW	1.55 m between B4 and the edge beam B3 (V6.4)
FF	8.95 0 – 13.31 6.5	solid 0.20	EW	maid wing, 4.36 m to the cut wall
FF	8.95 6.5 – 13.31 13	ribbed (hollow block) 0.30 (25+5)	EW	kitchen roof; suite bath sunken 0.15 (depressed zone x 10.3–13.311 y 6.5–10.45)
FF	8.95 13 – 13.31 19.25	ribbed 0.30	EW	dining roof; half on beam B5
FF	2.75 19.25 – 13.31 24	ribbed 0.30	NS	men's majlis roof, 4.75 m; thickened ribs B19–B21 under the FF walls; gym live load
roof	0 0 – 2.75 24	solid 0.20	EW	west band and lift / shaft heads
roof	2.75 0 – 4.25 24	solid 0.20	EW	riwaq / gallery line
roof	4.25 0 – 8.95 6.5	solid 0.20	EW	kids' room / family living: 4.7 m, no beam on y 3.0 (light partition)
roof	4.25 6.5 – 5.8 13	solid 0.20	EW	over the gallery strip: 1.55 m from R3 to the roof edge beam R11 on C21–C22 (X8; V6.4)
roof	4.25 13 – 8.95 19.25	solid, flat, no beams in the void 0.22	EW	the double-height living roof: 4.7 m R3 → R4, clean soffit
roof	8.95 0 – 13.31 19.25	ribbed 0.30	EW	bedroom 1 / suite: 4.36 m
roof	0 19.25 – 4.25 24	ribbed 0.30	NS	office roof: 4.75 m
roof	4.25 19.25 – 5.6 21.25	solid 0.20	NS	gym route
roof	4.25 21.25 – 13.31 24	solid 0.20	NS	gym roof: 2.75 m; its north edge R6 over the court
roof	11.59 19.25 – 13.31 21.25	solid 0.20	EW	treadmill arm roof

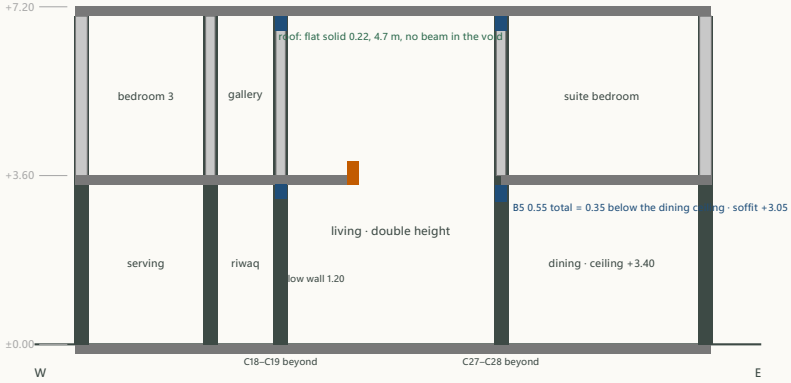
Sections · schematic, levels to confirm

A east–west through the riwaq, the courtyard and the colonnade (y 10) · B east–west through the living, the double-height void and the dining (y 16.5) · C north–south through the middle band (x 7.3)

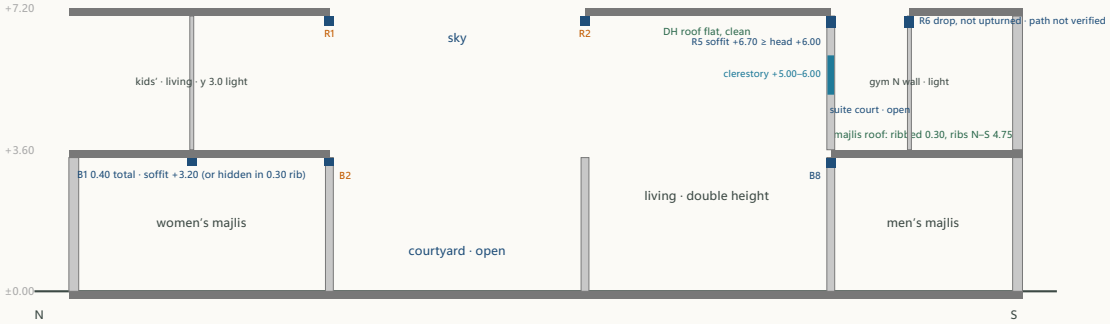
A · y 10: riwaq · courtyard · colonnade · kitchen



B · y 16.5: living (TV wall) · double-height void · dining · suite



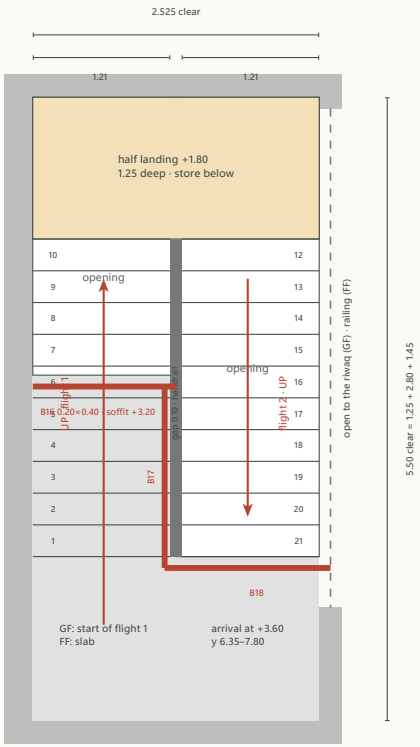
C · x 7.3: women's majlis · courtyard · double-height living · men's majlis roof · gym



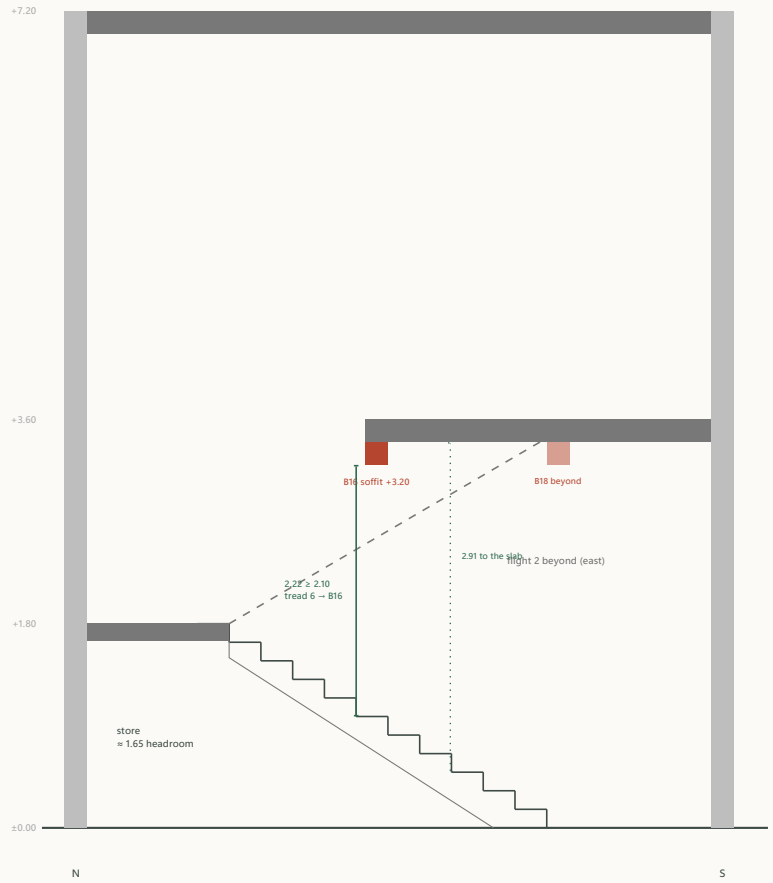
The stair · plan at the first-floor opening and section · all sizes assumed

well x 0–2.75 · y 2.2–7.9 (wall centre lines) · 22 risers × 0.164 = 3.60 m · goings 0.28 · two flights side by side · headroom measured to the beam soffits

stair plan at the first-floor slab · white = opening · grey = slab · north up



section along flight 1 (x ≈ 0.75), looking east



Numbers (assumed)

- **Well:** 2.75 × 5.70 m on wall centre lines = **2.525 × 5.50 m clear** (0.25 outer wall west, 0.20 walls north and south; the east side is open to the riwaq on the ground floor and a railing on the first floor).
- **Rise:** 3.60 m = **22 risers × 0.164 m**, 11 per flight; **goings 0.28 m**, 10 per flight = 2.80 m.
- **Plan:** half landing 1.25 m (y 2.30–3.55) at the north end, flights y 3.55–6.35, arrival 1.45 m (y 6.35–7.80) at the south end: **1.25 + 2.80 + 1.45 = 5.50 = the clear length. It fits.**
- **Flight clear width:** (2.525 − 0.10 gap) / 2 = **1.21 m each**, ≈ 1.13 m between a wall handrail (0.08) and the central handrail; ≥ 0.90 m (residential). Flight 1 west, up to the north; flight 2 east, up to the south.
- **Flight slabs:** 0.15 m solid, spanning landing to landing (2.80 m); the half landing on the stair walls (store under it, ≈ 1.65 m headroom: a store, not a room).

Opening in the first-floor slab (L-shaped, ≈ 8.2 m²)

- **Over flight 1 only beyond tread 6:** the slab above a tread must leave ≥ 2.10 m to the **beam** soffit (+3.20 with 0.40 total trimmers), so treads with a level ≤ 1.10 m may stay under the slab: treads 1–6 (tread 6 at +0.98 → 2.22 m). The opening edge B16 is at y 4.85 (beam 4.75–4.95), north of tread 6 (y 4.67–4.95) — tread 7 (y 4.39–4.67, +1.15) is under open sky.
- **Over flight 2 and the half landing:** fully open (x 1.39–2.65, y 2.30–6.35 and x 0.125–1.39, y 2.30–4.75).
- **Trimmers, 0.20 × 0.40 total, soffit +3.20:** B16 (y 4.85, from the west wall to B17), B17 (x 1.29, on flight 1's side of the split so it is never over flight 2, from B16 to B18), B18 (y 6.45, the arrival edge, from B17 to the riwaq line beam B10). B16 and B18 bear on B17: one beam-on-beam zigzag, normal at a stair well; the engineer takes the torsion.
- **Headroom:** 2.22 m at tread 6 to B16; 2.91 m at tread 3 to the slab; under B17 (over flight 1 treads 1–5, level ≤ 0.82) ≥ 2.38 m; flight 2 and the landing are under the roof (≥ 3.4 m). **No conflict with the reservation.**
- **Arrival:** flight 2 lands at +3.60 on the arrival strip y 6.35–7.80 inside the stair box, then the landing room (y 7.9–9.2) and the gallery.

Special zones · options compared (none imposed)

1 · The west projection: edge beam B3 on C21–C23 (spans 6.5 and 6.25 m)

	(a) drop beam	(b) upturned beam	(c) deeper slab edge
size (total, incl. the 0.20 slab)	0.25 × 0.50: 0.30 below the slab, soffit +3.10	0.25 × 0.50: 0.30 above the structural slab (top +3.90), i.e. ≈ 0.22–0.25 visible above a finished floor of 0.05–0.08; if the engineer needs 0.60 total, 0.40 above the slab	slab 0.35 over the 1.35 m strip, no beam
what you see	a 0.30 m downstand along the colonnade, 3.10 clear under it; the courtyard reads as a riwaq with a beam	a flush soffit at +3.40 over the colonnade; the upstand is the solid base of the gallery railing, the railing on it up to 1.10	flush soffit, a thicker slab edge 0.35 seen from the courtyard
structure	standard; the stiffest	standard; the slab hangs from the beam's bottom (stirrups) – routine	a 1.35 m wide, 0.35 deep band over 6.5 m: depth/span ≈ 1/18, deflection-sensitive, heavy
cost	lowest	≈ same formwork; saves the railing base	more concrete and steel
note	an acceptable alternative	**the owner's architectural preference**: flush colonnade soffit, railing base included; the engineer decides between (a) and (b)	only if a flush soffit AND an open railing are wanted

2 · The double-height living: a clean ceiling (preferred, not a rule)

- **Option 1 · flat solid roof slab 0.22 m spanning 4.7 m east–west** from the FF west rooms' wall (R3) to the suite wall (R4): no beam in the void, the two beams sit on the walls (invisible). A common span for a 0.22 solid slab (not calculated); cost ≈ a normal solid slab. **This study's preference.**
- **Option 2 · ribbed 0.30 slab, ribs east–west:** lighter and cheaper for the contractor, but a ribbed soffit (or a false ceiling) over the void; still no cross beam.
- **Option 3 · a cross beam on y 16.1 (mid-void) + 0.18 slab:** the cheapest in material, but a 0.45 m beam across the void, 7 m up, in the family's main room. Not recommended unless the engineer finds option 1 too deep.
- The void's walls: x 4.25 and x 8.95 are 7 m high but tied by the FF slab on their outer sides (the gallery / projection on the west, the suite floor on the east). The south wall (y 19.25, x 5.6–8.95, with the clerestory) is tied at +3.6 only by the men's majlis roof slab on its south face, and at +7.0 by the roof: **one tall wall, 7 m, tied on one side at mid-height** (X7).

3 · Living ↔ dining opening (x 8.95, 6.25 m) under the suite wall: beam B5

- A drop beam 0.25 × 0.55 total between C27 (8.95, 13.0) and C28 (8.95, 19.25); it carries the suite bedroom's west wall (3.4 m block), half the dining roof (ribbed, 4.36 m) and the roof beam R4 above the wall. 0.35 below the 0.20 slab: soffit +3.05 before finishes.
- **It can sit exactly on the ceiling-height change line:** the dining ceiling is at +3.40, the living has no ceiling, so the beam's 0.35 m below the dining ceiling becomes the step itself: one face of the double height. Headroom under it 3.05 m (≥ 2.40 required).
- **Alternative:** a wide-shallow beam 0.60 × 0.40 total (soffit +3.20) if the step should be thinner; or make the suite wall a light partition so the beam carries only the slab (0.25 × 0.45 total, soffit +3.15).

4 · Men's majlis roof (10.56 × 4.75 m) carrying the gym, the arm and the office

- **Span direction north–south, 4.75 m:** a ribbed (hollow-block) slab 0.30 m (25+5), ribs from the majlis north wall (B8) to the south wall (B9). **No column inside the majlis** (the room must stay free: 10.56 m long, seating at the east end).
- The first-floor walls **parallel to the ribs** (office wall x 4.25, gym route x 5.6, treadmill arm x 11.586) each stand on a **thickened rib / hidden beam** B19–B21 spanning the 4.75 m: direct and cheap.
- The gym's north wall **y 21.25 runs across the ribs** for 7.7 m with nothing below: (i) **a light partition** (100 mm block or dry wall, high windows onto the court) on the slab, designed for a partition load: **recommended**; (ii) a 200 mm block wall on a solid band 1.0 m wide in the ribbed slab; (iii) a drop beam the full 10.56 m (≈ 0.90 m deep, visible in the majlis): rejected.
- **Loads:** it is a **home gym** (light equipment, no dropped heavy weights, one treadmill): the engineer chooses a normal residential-gym live load as they judge (SBC 301 gives 5 kN/m² for gymnasia; what applies to a home gym is the engineer's call); the treadmill ≈ 1.5–2 kN dynamic (manufacturer's data); office 2.5–3 kN/m². **Vibration: not calculated here.** The natural frequency of the 4.75 m ribbed slab and its response under the treadmill arm must be checked by the engineer; the arm gets a floating floor and pads (DETAILS.md) in any case (X6).

5 · The stair (x 0–2.75, y 2.2–7.9)

- Drawn on the stair page: two flights of 1.21 m clear, 22 risers of 0.164 m, goings 0.28, a half landing 1.25 m deep at the north end, the arrival at the south end; the L-shaped opening of about 8.2 m² and the trimmers B16–B18 (0.40 total, soffit +3.20); headroom 2.22 m at the opening edge, measured to the trimmer soffit. **It fits the reservation.**

6 · Glazing and the grid

- **Courtyard glazing** (x 4.25 y 6.8–8.9 and 10.7–12.7; x 8.95 y 9.1–11.1): no column in the openings (C17, C18, C26, C27 at the corners); beams B4 / B6 above the heads. **Women's majlis south glazing** (y 6.5, x 5.1–8.1): C21 at x 5.8 sits in it → the glazing is 5.1–5.6 + 6.0–8.1 (accepted with the colonnade; V6.4). **Living glazing** (y 13.0, x 4.45–8.75): C22 at x 5.8 is the pier (the columns solution, tag v5.0). **Clerestory** (y 19.25, x 6.0–7.0 and 7.6–8.6, +5.00–6.00): C23 at x 5.6 and C28 at x 8.95 are clear of it; the roof beam R5 soffit +6.70 is 0.70 above the head. **Suite bath east window**, shortened by the owner to y 9.6–10.2: the clearance to C33 depends on the column's length along the wall, see X3.

7 · Lift shaft, air tower, roof openings

- **Lift (future, two stops):** RC walls W1 on x 0, y 0 (outer) and y 2.2, 0.20 m; the east wall x 2.25 blockwork with the future door reveal. The FF slab and the roof over the shaft (2.25 × 2.2) are **framed as openings now and closed with removable precast / lightweight panels** on hidden beams (B11 and the x 2.25 line), so the shaft opens later without cutting. Pit 1.2 m, isolated from the air tunnel.
- **Air tower** (8.95–10.594 × 0–1.0): rises through both slabs; the openings framed by B15 / R9 (y 1.0) and the walls x 8.95 (W3) and the cut wall (W4); the head 2.5 m above the roof in the same RC walls (studies/air_system.typ). **Below the slab the shaft walls continue to −1.95 as a dry base pit**, from which the small air duct (a sealed box 0.9 × 0.9, top −1.00, underside −1.90) leaves westward through the W3 line and runs south under the women's majlis on x 7.27 to the outlet chamber under the courtyard floor (x 6.75–7.95, y 6.6–7.6, floor −2.10, between the C21 and C26 pads); the outlet is a grille in the wall base at +0.2 … +0.8 through a plenum in the wall. The pit is the foundation of C24 / C30 and of the W3 / W4 ends; the women's majlis south wall bridges the duct mouth on a ground beam (air study items A1–A3). No water anywhere in the system.
- **Roof openings:** courtyard sky framed by R1, R2, R11 (west, on the colonnade, X8) and R4, the gallery strip roof between R3 and R11; suite court by R5, R6 and the FF walls x 5.6 / x 11.586 below the roof. West shaft (1.5 × 1.2) framed in B13 / R10.

Conflicts and open questions for the structural engineer

No.	State	Conflict / question
X1	accepted	Column **C22** (5.8, 13.0; V6.4, was 5.6) stands in the courtyard / living glazing (x 4.45–8.75): the glass is split 4.45–5.55 + 6.05–8.75 (the Ø0.40 stone sleeve); above the glass head the steel window W3 (+2.90 … +6.75) closes the void to the courtyard (V6.4, enclosure A). A decision of the columns solution (tag v5.0); keep the column round Ø0.30 and detail the glazing around it.
X2	accepted	Beam **B5** (x 8.95, 6.25 m, 0.25 × 0.55 total) shows 0.35 m below the dining ceiling, on the ceiling-height change line, soffit +3.05 before finishes. Acceptable: it is the step between the two heights. Alternative 0.60 × 0.40 total (soffit +3.20).
X3	owner decided · resolved (V6.3)	**C33 stays at (13.311, 10.45); the suite bath east window is 0.55 m, y 9.50–10.05 (V6.3).** Its frame (0.05) clears the assumed 0.25 × 0.60 column lying along the wall (y 10.15–10.75) by 0.05 m, and any smaller column by more: no clash for any column ≤ 0.60 along the wall. The GF kitchen window (y 8.7–10.1) is clear.
X4	rule	The gym roof's north edge **R6** (y 21.25) must be a **drop** beam, not upturned: the gym roof edge is a railing, not a parapet (solid top +7.20). An upstand of 0.40 would raise the solid top to +7.60 and cut the winter noon sun on the clerestory windows from ≈ 81 % to ≈ 40 %.
X5	NOT verified	**Proposed load path, NOT verified:** R6 (6.0 m) carried by the first-floor walls x 5.6 and x 11.586, which stand on thickened ribs of the men's majlis roof (B20, B21), not on columns: a transfer (wall on a beam). Reason: no column may stand inside the men's majlis. Proposal: build those two walls as RC walls 0.20 m (2 m long, slab to roof) so the path roof → wall → rib → GF walls is explicit. **The structural engineer must confirm the supports, the sizes and the deflection of R6, B20 and B21.** In the 3D model (phase 4) R6 and these two walls are to be shown as **unconfirmed** (own colour / hatch) until then. Loads: a home gym, see § 4.
X6	flag	The gym's north wall **y 21.25** (7.7 m) runs across the majlis roof ribs with nothing below: a **light partition** is recommended (100 mm block or dry wall; high obscured windows onto the court). If the owner wants a 200 mm wall, a solid band in the ribbed slab. Home gym (light equipment, one treadmill): live load as the engineer judges; the slab's vibration under the treadmill arm is **not calculated** and must be checked; floating floor and pads in the arm (DETAILS.md).
X7	flag	The living's south wall (y 19.25, x 5.6–8.95) is 7 m tall with the clerestory windows, tied at +3.6 only on its south face (men's majlis roof slab) and at +7.0 by the roof: its slenderness between columns C23 and C28 (3.35 m) is **not checked here**; the engineer confirms it, with RC stiffeners at the window jambs if needed.
X8	owner decided	**Colonnade columns C21–C23 continue to the roof** inside the gallery railing line (a two-storey colonnade, slender round columns ≈ Ø0.30 assumed). The roof over the gallery strip (1.55 m, y 6.5–13.0; V6.4) rests on a roof edge beam **R11** (x 5.8, 0.25 × 0.45 total) between C21 and C22: no roof cantilever. C22 also takes the west end of R2, C23 stands under R5. The engineer sizes the columns for two storeys and the two roof beams.
X9	missing	**No soil report, no soil data:** footing type and depth (pads ≈ 1.5 × 1.5 assumed, underside ≈ −1.50), the lift pit (1.2 m), and the **air duct** (studies/air_system.typ v2): a small sealed box 0.9 × 0.9, top −1.00 / underside −1.90, leaving the tower base pit (to −1.95) through the W3 / C24 line and running under the women's majlis to the outlet chamber under the courtyard floor (floor −2.10) between the C21 and C26 pads; the pit as the foundation of C24 / C30 and the W3 / W4 ends, the ground beam of the majlis south wall over the duct mouth: engineer (air items A1–A3).
X10	flag	**Suite walls over the kitchen** (x 10.3, y 10.45, x 12.211, y 8.0, y 11.9, y 13.9): light partitions on the ribbed slab, or hidden beams under the bath walls that carry pipes; the **suite bath floor is sunken** (depressed slab ≈ 0.15 over x 10.3–13.311, y 6.5–10.45) with nothing in the kitchen ceiling: a depressed-slab detail inside a ribbed bay.
X11	flag	**Floor-to-floor 3.60** with a 0.20 slab and beams of 0.40–0.55 total depth gives 3.05–3.20 m clear under the beams before finishes (B4 soffit +3.10 over the riwaq glazing, B5 +3.05 at the dining step, B1 +3.20 in the women's majlis). Acceptable (≥ 2.40). **Owner decision:** B1 is a drop beam hidden by a gypsum ceiling (≈ +3.05); the general rule is that drop beams in finished rooms are hidden by gypsum ceilings, which also carry the hidden AC and lighting.
X12	open	**Beam B4** (x 4.25, 6.5 + 6.25 m) carries the first-floor bedrooms' east wall over the riwaq glazing (y 6.8–8.9, 10.7–12.7) and the low wall: with 0.25 × 0.50 total (soffit +3.10) the riwaq glazing heads are at ≤ +3.10 before finishes, and the dining-side heads under B5 at ≤ +3.05; confirm the window heights with the architect.
X14	owner decided (V6.4)	**The kitchen's north wall moves to y 7.0** (maid room to the municipal minimum): a **non-load-bearing block partition** on a thickened slab / ground beam (it crosses the malqaf duct at x 10.25 on a ground beam, air study A3). The structural grid is unchanged: the y 6.5 line keeps **B6** as a **drop beam** (0.25 × 0.45 total assumed) from C26 to the east wall, inside the laundry / maid room ceiling behind the gypsum ceiling, carrying the first-floor bedroom 1 / suite wall; the FF slab bays stay on the grid. W5 is dropped from the RC-wall proposal (X13).
X15	owner decided (V6.4)	**Courtyard enclosure, option A** (studies/courtyard_enclosure.typ): the west projection is **1.55 m** (edge x 5.8); B3, R11 and C21–C23 move 0.20 m east, spans and X8 unchanged. The gallery edge, the family living's edge and the living's upper courtyard side (y 13.0, +2.90 … +6.75) are closed with black steel windows W1 / W2 / W3; W1 sits on the B3 upstand (its sill), heads at the R1 / R2 / R11 soffit +6.75; a 0.25 m fascia keeps the head line under the roof slab over the void. W3 needs a steel transom at +3.82 tied to C22 and the kitchen wall; the window loads are small. The gym route wall (B20) stays on x 5.6: a 0.20 m nib at y 19.25 (to follow to x 5.8 in the L study if the owner wants one line).
X13	open	**Which walls are RC:** this study proposes W1–W4 (lift, shaft, laundry wall, cut wall) as RC 0.20 m for stiffness (W5, the kitchen north wall, dropped in V6.4), plus the two gym walls of X5. The engineer confirms the lateral system (frame + walls) under SBC 301 and the wind / seismic case for Taif.
X14	open	**Slab types:** ribbed 0.30 (hollow block) for the 4.36–4.75 m bays and the men's majlis roof; solid 0.20 elsewhere and 0.22 flat over the double-height living. The engineer may unify (all ribbed, or all solid 0.20–0.25) for buildability; the clean soffit over the void is the only architectural wish.
X15	open	**Stair** (drawn on the stair page): flight widths 1.21 m clear, 22 risers × 0.164, the L-shaped opening and the trimmers B16–B18 with their torsion, the store under the half landing (≈ 1.65 m headroom), handrails; the arrival strip y 6.35–7.80 at the first floor.

Rules of thumb used (all assumed): beam depth = TOTAL depth incl. the slab, drop beams ≈ span / 12, hidden beams = slab depth, solid one-way slabs ≈ span / 24–28 (0.20 for ≤ 5 m), hollow-block ribbed slabs 0.30 (25 + 5) for 4–6 m, flat solid roof 0.22 for 4.7 m, columns 0.25 × 0.50–0.60 in the walls (two storeys, light loads), colonnade Ø0.30, pad footings 1.5 × 1.5 m. Loads: SBC 301 (residential 2 kN/m², gym: a home-gym live load as the engineer judges, roof 1 kN/m² + plant). Nothing here replaces the structural design.

✓ **Preliminary structural study · 36 columns · 21 first-floor beams · 11 roof beams · 17 items for the engineer**